

DTSA 5012

ANOVA and Experimental Design

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Applied Scientist (57.5 / 100)	49.1 / 100
SPECIALIZATION Statistical Modeling for Data Science Applications		

49

CURRENT AVG SCORE
across 6 job roles

79

POTENTIAL SCORE
with recommended additions

+40

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduction to the analysis of variance (ANOVA), analysis of covariance (ANCOVA), and experimental design. ANOVA and ANCOVA, presented as a type of linear regression model, provide mathematical basis for designing experiments for data science applications. Emphasis placed on important design-related concepts, such as randomization, blocking, factorial design, and causality. Attention will also be given to ethical issues raised in experimentation.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	57.5	94.0	+36.4	HIGH
ML Engineer	54.6	80.7	+26.2	MEDIUM
Machine Learning Engineer	51.9	91.9	+40.0	HIGH
MLOps Engineer	45.8	74.6	+28.8	MEDIUM
AI Engineer	42.5	75.0	+32.5	HIGH
Data Scientist	42.4	59.4	+17.0	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

✓ This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

Applied Scientist	57.5 → 94.0 (+36.4 pts)
computational problems models learning development testing techniques machine	
ML Engineer	54.6 → 80.7 (+26.2 pts)
media learning machine models problems global tools	
Machine Learning Engineer	51.9 → 91.9 (+40.0 pts)
predictive learning tools exploratory development models machine information	
MLOps Engineer	45.8 → 74.6 (+28.8 pts)
learning machine media tools models development error	
AI Engineer	42.5 → 75.0 (+32.5 pts)
perception models learning problems machine information tools make	
Data Scientist	42.4 → 59.4 (+17.0 pts)
behaviors problems	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.